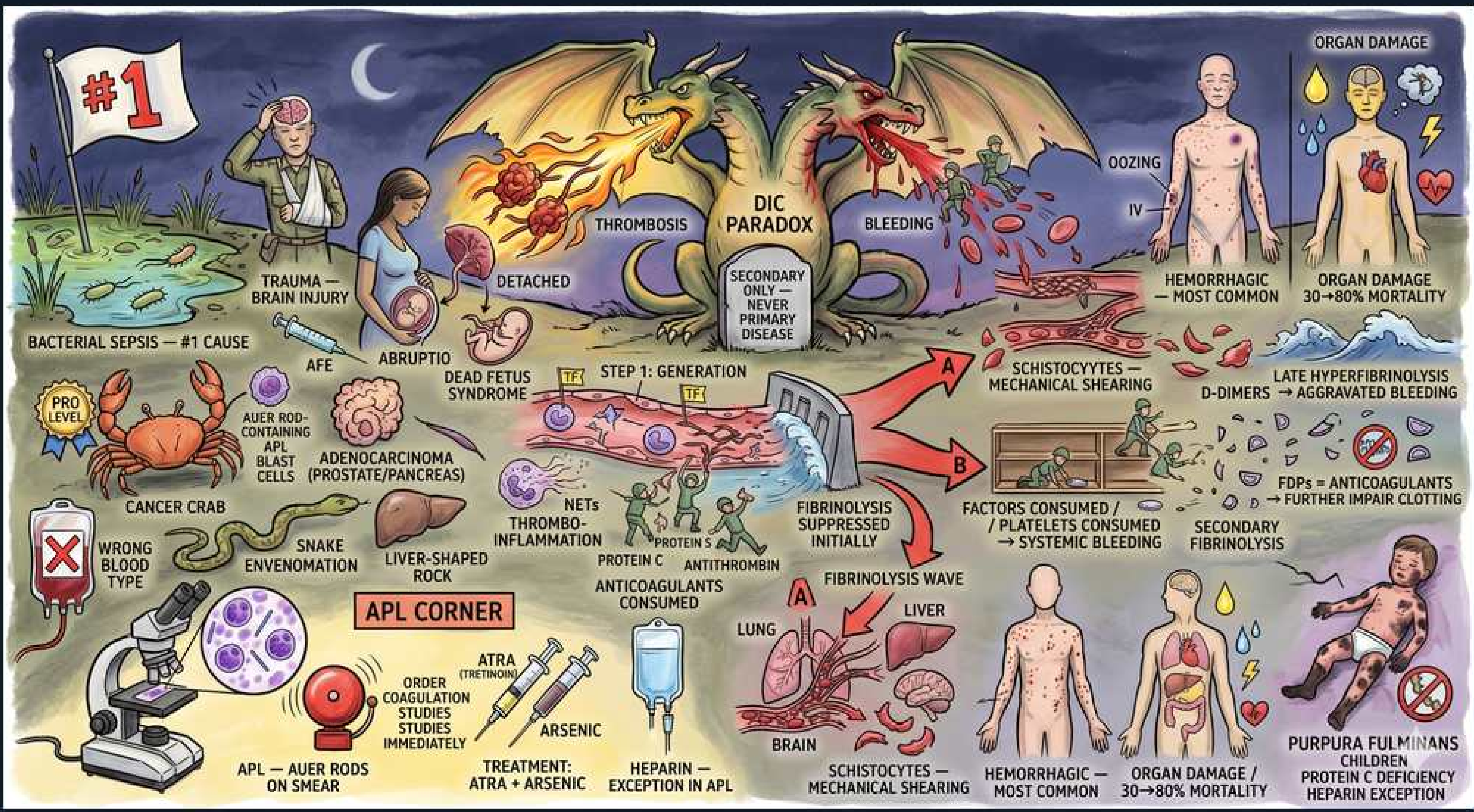
Coagulation — DIC (mechanism, triggers, consumption)



1. DIC paradox: clot + bleed

WHAT YOU SEE

Two opposed dragons center-stage labelled THROMBOSIS and BLEEDING — the DIC paradox of clotting and hemorrhage happening at once

WHAT IT ENCODES

DIC is a systemic, uncontrolled activation of the coagulation cascade: widespread tissue-factor exposure drives explosive thrombin generation that lays down fibrin microthrombi throughout the microvasculature (causing ischemic organ damage), while simultaneously consuming platelets, fibrinogen and clotting factors faster than they are made, producing diffuse bleeding. The two dragons capture the central paradox — the same process causes both thrombosis and hemorrhage at the same time. Secondary fibrinolysis then chews up the clots, adding fibrin degradation products (FDPs) that further impair hemostasis.

BOARD TRAP

The classic board trap is framing DIC as 'just a bleeding disorder' or 'just a clotting disorder' — it is a consumptive coagulopathy that is BOTH at once; microthrombi cause the organ failure that drives the 30–80% mortality even when bleeding dominates the bedside picture.

2. Always secondary

WHAT YOU SEE

A center banner 'secondary only — never primary disease' — DIC is always a downstream syndrome, never a stand-alone diagnosis

WHAT IT ENCODES

DIC is never a primary disease — it is always secondary to an underlying trigger (sepsis, obstetric catastrophe, malignancy, trauma/burns, snakebite, transfusion reaction). The single most important therapeutic step is identifying and treating that trigger; blood-product support is purely adjunctive. Think of DIC as a lab/clinical syndrome, not a diagnosis in itself.

BOARD TRAP

There is no such thing as 'primary DIC' — if a question offers DIC as a standalone diagnosis without a precipitant, look for the hidden cause (occult sepsis, retained dead fetus, APL). Correcting the numbers without treating the cause is a wrong answer.

3. Sepsis — #1 cause

WHAT YOU SEE

A big '#1' flag over bacterial-sepsis imagery — sepsis is the single most common DIC trigger

WHAT IT ENCODES

Bacterial sepsis is the most common cause of DIC overall. Endotoxin (LPS) from gram-negative organisms and cytokines (TNF-α, IL-1, IL-6) induce massive tissue-factor expression on monocytes and endothelium, igniting the cascade. Meningococcemia (Neisseria meningitidis) is the prototypical sepsis-DIC that produces purpura fulminans and Waterhouse-Friderichsen adrenal hemorrhage.

BOARD TRAP

Sepsis is the #1 trigger — ahead of malignancy and obstetric causes — and gram-negative organisms are the classic culprits; a hypotensive febrile patient with new thrombocytopenia and oozing IV sites is DIC until proven otherwise.

4. Obstetric triggers

WHAT YOU SEE

Left 'AFE, abruptio, dead fetus syndrome' fetus icons

WHAT IT ENCODES

Obstetric catastrophes are leading DIC triggers because placenta and amniotic fluid are extremely tissue-factor rich. The classics: amniotic fluid embolism (sudden cardiopulmonary collapse + DIC during labor), placental abruption, retained dead fetus syndrome (chronic low-grade DIC over weeks), and severe preeclampsia/HELLP. Sepsis from septic abortion or endometritis is another route.

BOARD TRAP

AFE and abruption cause acute fulminant DIC, whereas a retained dead fetus causes a slow, compensated/chronic DIC — recognize the retained-products scenario as the answer when a woman has been carrying a nonviable fetus for weeks with falling fibrinogen.

5. Malignancy (adenoCA, APL)

WHAT YOU SEE

Crab 'cancer crab' and 'adenocarcinoma prostate/pancreas' plus blast cells

WHAT IT ENCODES

Malignancy is a major chronic-DIC trigger. Mucin-secreting adenocarcinomas (pancreas, prostate, GI, lung) release tissue factor and procoagulant mucins, classically producing chronic compensated DIC and migratory thrombophlebitis (Trousseau syndrome). Acute promyelocytic leukemia (APL, AML M3) is the most feared — the malignant promyelocytes are packed with procoagulant/fibrinolytic granules (Auer rods) that trigger a severe HEMORRHAGIC DIC.

BOARD TRAP

APL classically presents with life-threatening bleeding from DIC at diagnosis — fatal intracranial/pulmonary hemorrhage is a board killer; the move is to start ATRA immediately on clinical suspicion, before genetic confirmation, rather than wait.

6. Tissue factor / generation

WHAT YOU SEE

A center 'STEP 1: generation' panel showing protein C / antithrombin being used up — the initiating thrombin burst consumes natural anticoagulants

WHAT IT ENCODES

The initiating step is excess tissue factor (TF) exposure activating the extrinsic pathway → a thrombin burst → fibrin everywhere. This thrombin storm simultaneously consumes the body's natural anticoagulants — antithrombin, protein C and protein S — removing the brakes and amplifying the thrombosis. Loss of protein C is what links severe DIC to purpura fulminans.

BOARD TRAP

DIC begins with thrombin OVERgeneration and consumption of natural anticoagulants — the early phase is prothrombotic; the bleeding phenotype emerges only after factors/platelets are exhausted. Don't assume the first event is bleeding.

7. Factors & platelets consumed

WHAT YOU SEE

A center-right panel 'factors consumed / platelets consumed -> systemic bleeding' — the consumptive coagulopathy exhausting hemostasis

WHAT IT ENCODES

As clotting runs unchecked, platelets, fibrinogen and coagulation factors are consumed faster than the marrow and liver can replace them — hence 'consumptive coagulopathy.' This yields the diagnostic quartet: thrombocytopenia, low/falling fibrinogen, prolonged PT and aPTT. Factor VIII is also consumed and falls — a key point separating DIC from liver disease.

BOARD TRAP

The thrombocytopenia and hypofibrinogenemia here are from CONSUMPTION, not marrow failure or impaired synthesis — a peripheral smear showing schistocytes plus a high D-dimer distinguishes consumptive DIC from production problems like leukemia infiltration or B12 deficiency.

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8. Secondary fibrinolysis / D-dimer

WHAT YOU SEE

A center-right 'fibrinolysis wave' releasing D-dimers/FDPs — secondary plasmin breakdown of the fibrin clot

WHAT IT ENCODES

Widespread fibrin clot triggers a compensatory plasmin response (secondary fibrinolysis) that lyses cross-linked fibrin, releasing D-dimer and fibrin degradation products (FDPs). D-dimer is the single most sensitive lab marker of DIC. FDPs are themselves anticoagulant — they impair fibrin polymerization and platelet function, worsening the bleeding.

BOARD TRAP

A HIGH D-dimer reflects breakdown of cross-linked fibrin (true clot lysis) and is sensitive but NOT specific — it also rises in PE/DVT, surgery, pregnancy and trauma; conversely, isolated high FDPs with a normal D-dimer suggests PRIMARY fibrinolysis, not DIC.

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9. Schistocytes — MAHA

WHAT YOU SEE

'Schistocytes — mechanical shearing' fragmented red cells

WHAT IT ENCODES

Fibrin strands strung across small vessels shear passing red cells into fragments (schistocytes/helmet cells), producing a microangiopathic hemolytic anemia (MAHA) with elevated LDH, low haptoglobin and indirect hyperbilirubinemia. Schistocytes on smear are a hallmark of microvascular thrombosis.

BOARD TRAP

Schistocytes are NOT specific to DIC — they also appear in TTP, HUS, HELLP and malignant hypertension; the way to nail DIC is the combination of schistocytes WITH prolonged PT/aPTT, low fibrinogen and high D-dimer (in TTP/HUS the PT, aPTT and fibrinogen are normal).

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10. Prolonged PT and aPTT

WHAT YOU SEE

Top-right oozing patient and organ-damage panels — the consumptive phenotype where both PT and aPTT prolong as all factors fall

WHAT IT ENCODES

The full consumptive lab signature is prolongation of BOTH PT and aPTT, thrombocytopenia, low fibrinogen and elevated D-dimer/FDPs. Both arms prolong because factors of the extrinsic and intrinsic/common pathways are all being consumed. Compare to hemophilia (isolated long aPTT) or warfarin/early liver disease (PT first).

BOARD TRAP

DIC prolongs BOTH PT and aPTT — an isolated prolonged aPTT with normal PT points to hemophilia or a factor XII/lupus-anticoagulant issue, not DIC; the global derangement of all values is the giveaway.

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11. Hemorrhagic most common

WHAT YOU SEE

A right-side bleeding/oozing patient with '30-80% mortality' organ-damage note — hemorrhage is the most common presentation

WHAT IT ENCODES

Bleeding/oozing from venipuncture sites, surgical wounds, GI tract and mucosa is the most common clinical presentation of acute DIC. But beneath the obvious hemorrhage, microthrombi silently infarct kidneys, lungs (ARDS), liver and CNS, driving a mortality of roughly 30–80% depending on the trigger. Acrocyanosis and digital gangrene reflect the thrombotic component.

BOARD TRAP

Do not be lulled by the bleeding — the thrombotic organ injury is what kills; a DIC patient may need supportive products for bleeding while the real prognosis hinges on reversing organ ischemia and the underlying trigger.

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12. Purpura fulminans

WHAT YOU SEE

Bottom-right infant legs 'purpura fulminans, protein C deficiency'

WHAT IT ENCODES

Purpura fulminans is fulminant dermal microvascular thrombosis producing retiform purpura, hemorrhagic skin necrosis and limb gangrene. It is driven by acute depletion of protein C (and S) and is classically seen in meningococcemia, severe sepsis-DIC, and in neonates with homozygous protein C deficiency. Treatment includes protein C concentrate/FFP plus treating the cause.

BOARD TRAP

Purpura fulminans is a THROMBOTIC skin manifestation (microvascular occlusion with necrosis), not simple bruising — recognize it in a septic child with rapidly spreading dark purpuric necrotic patches and think meningococcemia / protein C deficiency.

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13. APL: heparin exception, ATRA

WHAT YOU SEE

The bottom-left APL corner with ATRA + arsenic vials and a 'heparin exception in APL' tag — APL is the DIC where the leukemia itself drives the bleeding

WHAT IT ENCODES

In APL (AML M3, t(15;17), PML-RARA), the cure-defining move is urgent all-trans retinoic acid (ATRA), which differentiates the promyelocytes and rapidly shuts down the DIC; arsenic trioxide is added for definitive therapy. Aggressive product support (platelets to keep >30–50k, cryoprecipitate for fibrinogen >100–150, FFP) is given because the DIC is hemorrhagic. APL is the textbook situation where the underlying disease itself is the DIC driver.

BOARD TRAP

Start ATRA on clinical suspicion of APL — do NOT wait for cytogenetic confirmation, because untreated APL-DIC kills via intracranial/pulmonary hemorrhage within days; also remember ATRA carries its own differentiation syndrome risk (treat with steroids).
